PACHETUL GAUSSIAN

$ORIGIN := 1$

$m := 9.1 \cdot 10^{-31}$

$h := 6.625 \cdot 10^{-34}$

$hp := \frac{h}{2 \cdot \pi}$

$dt := 10^{-16}$

$dx := \frac{10^{-8}}{500}$

$a1 := 1 + dt \cdot \frac{i \cdot hp}{2 \cdot m \cdot dx^2}$

$a2 := -dt \cdot \frac{i \cdot hp}{4 \cdot m \cdot dx^2}$

$b1 := 1 - dt \cdot \frac{i \cdot hp}{2 \cdot m \cdot dx^2}$

$b2 := dt \cdot \frac{i \cdot hp}{4 \cdot m \cdot dx^2}$

$i := 1 .. 500$

$k := 1 .. 500$

$A_{i,k} := a1 \cdot \delta(i,k) + a2 \cdot \delta(i,k-1) + a2 \cdot \delta(i,k+1)$

$B_{i,k} := b1 \cdot \delta(i,k) + b2 \cdot \delta(i,k-1) + b2 \cdot \delta(i,k+1)$

Scriem functia de unda la momentul initial

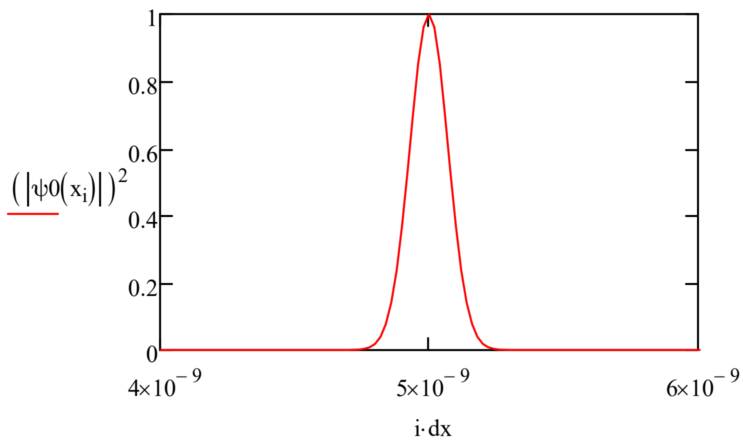
$\sigma := 10^{-10}$

$\lambda := 5 \cdot 10^{10}$

$x0 := 0.5 \cdot 10^{-8}$

$$\psi 0(x) := \exp\left[\frac{-(x - x0)^2}{2\sigma^2}\right] \cdot \exp(i \cdot \lambda \cdot x)$$

$x_i := i \cdot dx$



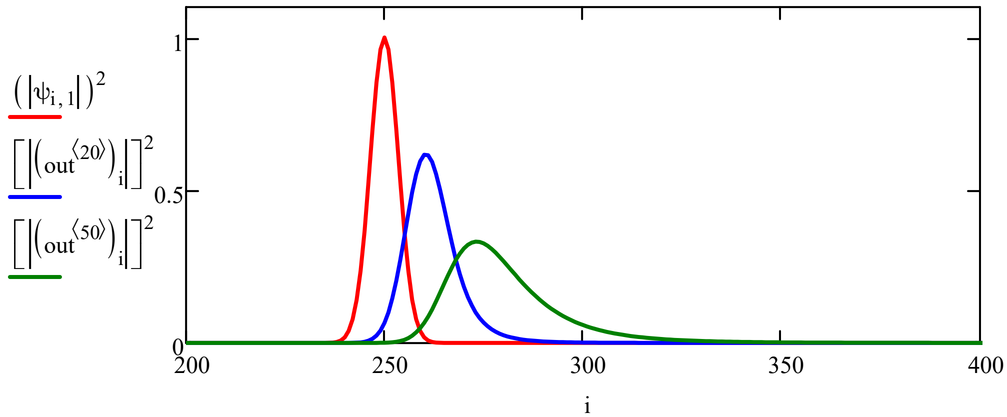
Rezolvam sistemul de ecuatii

$\psi_{i,1} := \psi 0\left(x_i\right)$

$V^{\langle 1\rangle} := B \cdot \psi^{\langle 1\rangle}$

$S^{\langle 1\rangle} := \mathrm{lsolve}\left(A, V^{\langle 1\rangle}\right)$

$$
out := \left|\begin{array}{l}
\text{for } n \in 2..200 \\
\quad \left|\begin{array}{l}
\psi^{\langle n\rangle} \leftarrow S^{\langle n-1\rangle} \\
V^{\langle n\rangle} \leftarrow B \cdot \psi^{\langle n\rangle} \\
S^{\langle n\rangle} \leftarrow \mathrm{lsolve}\left(A, V^{\langle n\rangle}\right)
\end{array}\right. \\
S
\end{array}\right.
$$

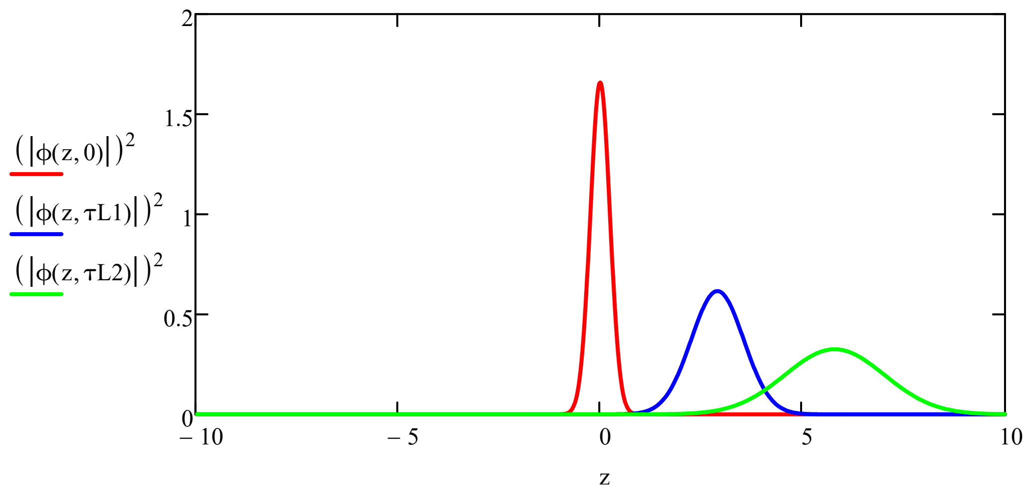


DISPERSIA PACHETULUI

$p0 := 10 \cdot hp$ $\qquad \tau L2 := 5000 \qquad \tau L1 := 2500$

$\alpha := 0.001 \cdot \frac{m}{hp}$

$$\phi(x,\tau) := \sqrt[4]{\frac{\alpha}{\pi}} \cdot \frac{1}{\sqrt{1 + i \cdot \frac{\alpha \cdot hp \cdot \tau}{m}}} \cdot e^{\frac{1}{2 \cdot \alpha \cdot hp^2} \cdot \left[\frac{(p0 + i \cdot hp \cdot \alpha \cdot x)^2}{1 + i \cdot \frac{\alpha \cdot hp \cdot \tau}{m}} - p0^2\right]}$$



Animatie:
====> tools
====> animation
====> record

Obs: contorul pt animatie
este variabila FRAME